Low Power 4-Bit Full Adder Cells in Subthreshold Regime

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***Abstract*— In this work, low power 4-bit adder cells based on subthreshold logic in nanometer CMOS technology is presented. Twelve different subthreshold 1-bit full adders are used for designing 4-bit adders in two different topologies. These topologies are Carry-Ripple and Carry Look-Ahead adders. To compare these adders, HSpice simulations in 65nm and 90nm standard CMOS technologies are used. Different performance parameters of the adder including power, delay, power-delay product and energy-delay product as a function of supply voltage, transistor sizing, frequency, technology, and temperature are investigated. The best adder with the smallest power and power-delay product and the highest achievable frequency is determined.**

Keywords: Subthreshold logic, Ultra low power CMOS, High performance, Carry-Ripple adder, Carry Look-Ahead adder.

I. INTRODUCTION

In recent years, the growth of portable devices such as notebook and laptop computers has caused many efforts for optimizing energy or power for a given performance. In some applications, such as wearable wrist-watch computers, medical devices like hearing aids, pace-makers, defibrillators and self-powered devices, ultra-low power consumption is the primary requirement while the performance is of a secondary importance. For these applications, operating the transistors of digital logic in the subthreshold region has been proposed [1]. In this regime, the supply voltage (V_{DD}) is less than the transistor threshold voltage (V_{th}). The operating current in the subthreshold regime is [2]

$$I_{DS} = I_S e^{\frac{V_{GS}+\eta V_{DS}-V_{th}}{nV_T}}\left(1-e^{-\frac{V_{DS}}{V_T}}\right) \quad (1)$$

Where

$$I_s = \mu_0 C_{ox}\frac{W}{L}(n-1)V_{th}^{\ 2} \quad (2)$$

The parameters in (1) and (2) are as defined as:

V_T	Thermal voltage ((KT/q) = 26 mV at 300°K)
n	The subthreshold slope parameter ($n = 1+C_{DEP}/C_{ox}$)
$\mu_{n(p)}$	The carrier mobility for n-(p-) channel device.
C_{ox}	The oxide capacitance.
C_{DEP}	The depletion capacitance.
W	Effective width of the channel.
L	The Length of the channel.

This current is known as the leakage current which restrict the maximum circuit performance, but because of using lower supply voltage, a circuit operating in subthreshold region has been shown to consume orders of magnitude less power than the circuits which biased in strong inversion region.

The adder is one of the basic elements in many VLSI systems. Also addition is used in many other arithmetic operations such as subtraction and multiplication. Thus, lowering the power consumption of the adders is an important design objective for low power digital designs [3]. One of the best methods of reducing full adder power consumption is obtained by using subthreshold style design. Some different designs of 1-bit full adder cell operating in subthreshold region have been proposed [4]–[7]. In [4] fourteen 1-bit full adder cells based on dividing the cell into three different blocks, operating in the subthreshold region are investigated and their characteristics as functions of different parameters such as supply voltage, sizing of transistor and technology are studied. The results show that the best determined full adder cell in subthreshold regime can differ from the strong inversion region. Six 1-bit full adder topologies operating in subthreshold region are simulated and analyzed in [5] based on comparing Min3-based full adder structures (Stacked, Mirrored and IJCNN) to CMOS full adder topologies and the advantages and disadvantages of each full adder cell operating in subthreshold region is presented. In [6] Subthreshold Domino full adder circuit is presented. The simulation results show that this structure has lower power consumption, smaller

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area, better speed and higher noise margin in comparing to subthreshold CMOS adder. A Differential Cascode Voltage Switch Pass Gate (DCVSPG) Subthreshold full adder is presented in [7]. This adder has 66% improvement in EDP in compare to CMOS full adder. This improvement in performance is due to single PMOS stack depth, positive feedback and reduced NMOS stack depth.

No multi-bit subthreshold adder has been reported yet. Besides that the proposed adders in [5] have the power consumption or performance or driving capability limitations, while the domino adder in [6] has related dynamic back drawn such as charge sharing, clock feed-through and etc. especially in multi-bit systems. In [7], the maximum frequency is 2.5 MHz that there is a concern about multi-bit systems to be reduced. In this work, two adder topologies are used to design the subthreshold 4-bit adders to achieve lower power consumption and higher performance in comparison with previous works, while these adders does not involve with the problems of mentioned adder in [6]. The first topology is the Carry-Ripple adder (CRA) which is constructed by cascading simple 1-bit full adders. For this adder, the carries in different bit positions are generated sequentially where each carry bit is dependent on the carry bit of the previous stage. This sequential process increases the delay of the adder as the number of bits increases [8]. In most of VLSI systems, the adder is part of the critical path that affects the performance of the system. For this reason improving the performance of the building blocks of the binary adders is an important consideration [3]. A more complex binary adder that has a better performance relative to the first topology (non-tree based CRA adders) is the Carry Look-Ahead adder (CLA). In this adder where all the carries are generated in parallel, the problem of the delay increase due to the carry propagation does not exist [8].

In this paper, first twelve subthreshold 1-bit full adders [4] are used for designing the 4-bit Carry-Ripple adder. For improving the speed of the subthreshold adder, the 1-bit full adder blocks are used for designing a subthreshold 4-bit Modified Carry Look-Ahead (MCLA) adder proposed in [9]. The paper is organized as follows. The blocks of a 1-bit full adder cell are discussed in Section 2 while in Section 3, the 1-bit full adders are used for implementing the subthreshold 4-bit Carry-Ripple adder. Section 4 contains the implementation of the 4-bit Modified Carry Look-Ahead (MCLA) adder. Finally, the summary and conclusion are given in Section 5.

II. THE BLOCKS OF 1-BIT FULL ADDER

The 1-bit full adder cell has three inputs (*A, B,* C_{in}) and two outputs (*sum,* C_{in}). The basic relations between these inputs and outputs are

$$sum = (A \oplus B) \oplus C_{in} \tag{3}$$

$$C_{out} = A.B + C_{in}.(A \oplus B) \tag{4}$$

A 1-bit full adder can be divided into three basic building blocks. The first block is used for implementing XOR between *A* and *B* inputs and its complement. The second block calculates *sum* output using the outputs of the first block and C_{in} while the third block is used to calculate the output *carry* (C_{out}) using the outputs of the first block, *A*, and C_{in}. Fig. 1 shows these building blocks and their relations. The Boolean expression that shows the relation between the inputs and outputs of these blocks are as follows [3]:

- Block 1:

$$H = A \oplus B \tag{5}$$

- Block 2:

$$sum = H \oplus C_{in} = H.\overline{C}_{in} + \overline{H}.C_{in} \tag{6}$$

- Block 3:

$$C_{out} = A.\overline{H} + C_{in}.H \tag{7}$$

The first block generates XOR and XNOR functions between *A* and *B* inputs. Three different structures for the first block are shown in Fig. 2. For having XNOR gate we can generate XOR and then use an inverter to generate the XNOR corresponding to Fig. 2(a) and (b) or we can have both outputs generated simultaneously with the structure of Fig. 2(c). The structure in Fig. 2(b) has the lowest transistor count and consequently the lowest power consumption.

The second block generates the sum output using the outputs of the first block and C_{in}. Since it is an XOR function, the same structure as the first block can be used for this block. Four different structures for the second block are shown in Fig. 3. The structure in Fig. 2(b) has no ground- or power-supply rails so there is no short-circuit current. Also, it has the lowest transistor count and as a result it has the lowest power consumption.

The third block generates C_{out} using *A*, C_{in}, and the outputs of the first block. This block is a multiplexer passing either *A* or C_{in} , according to the value of *H*. For this block, only one structure can implement the function by four transistors using transmission gates. It can provide enough driving power for loading the following gates. This structure is shown in Fig. 4.

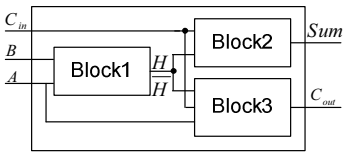


Figure 1. The building blocks of 1-bit full adder cell [3].

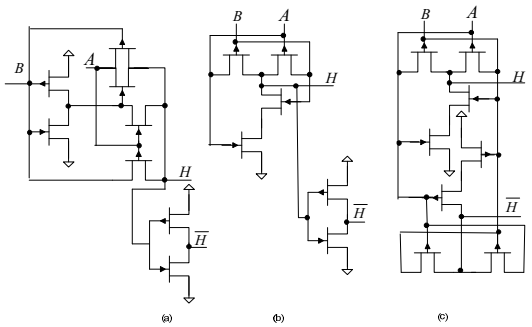


Figure 2. Three different structures for *Block 1* [3].

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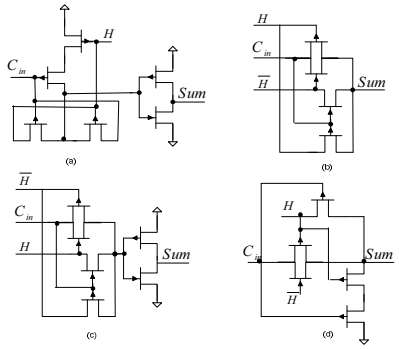


Figure 3. Four different structures for *Block 2* [3].

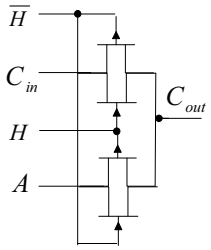


Figure 4. The structure of *Block 3* [3].

Twelve 1-bit full adder cells can be constructed using the various structures of each block [3]. These adder cells can work properly in the subthreshold region [4]. Note that the first letter of the adders name refers to the first block structure and the second letter refers to second block structure.

III. 4-Bit Carry-Ripple Adder Cells

The twelve 1-bit subthreshold full adder cells are used to build the 4-bit subthreshold Carry-Ripple adder. A *k*-bit Carry-Ripple adder is constructed using *k* 1-bit full adders. In this structure, the carry bit is propagated among the 1-bit full adders to produce the carry-out bit [10]. Fig. 5 shows a 4-bit Carry-Ripple adder.

The 4-bit full adders are simulated using HSpice in 65nm and 90nm standard CMOS technologies [11]. The transistor sizing in subthreshold may have affected the power consumption. Here, we consider equal *(W/L)* ratio of 6 is for all nMOS and pMOS transistors except for the pMOS transistors of the inverters which we use the ratio of 9. The simulations were performed at different frequencies (ranging from 1 KHz to 7 MHz) and supply voltages (ranging from 0.15V to 0.3V). We assumed the outputs load capacitance of 100fF. For having more realistic simulation condition, buffers are used before all the inputs.

The input waveforms discussed in [12] contain all the possible transitions from one input combination to another (56 patterns). We repeat the waveforms at random to generate the 4-bit inputs.

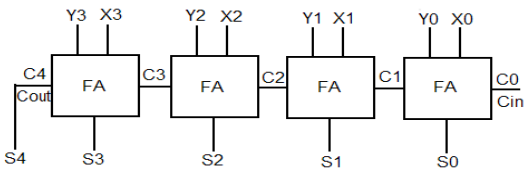


Figure 5. 4-bit Carry-Ripple adder [10].

The propagation delay in all paths, $(a_0 \rightarrow sum_{k-1})$, $(b_0 \rightarrow sum_{k-1})$, $(c_0 \rightarrow sum_{k-1})$, $(c_0 \rightarrow c_k)$, $(c_0 \rightarrow c_k)$ and $(c_0 \rightarrow c_k)$ are measured and then the largest one is reported as the propagation delay of the circuit.

The simulation results for the 4-bit CRA adders based on the twelve 1-bit full adders discussed in the previous section, and the static CMOS full adder cell and Transmission-Gates CMOS (TG-CMOS) full adder cell at frequency of 1 MHz and the supply voltage of 0.2V are shown in Table I. The results are sorted by the power consumption. These results show that the adders with *bb* and *bd* cells with the lowest number of transistors have the least power consumptions, and the adder with *ca* cells that generates XOR and XNOR outputs simultaneously, have the least delay, power-delay and energy-delay products.

The power consumption, power-delay and energy-delay product of the CRA with the *bd* full adder cell (which has the least power) as a function of the supply voltage at the frequency of 1MHz are shown in Fig. 6. By increasing the supply voltage, the power consumption of the adder increases while its delay decreases. This decrease is more significant and as a result the power-delay and energy-delay product of the adder decrease.

Then, the 4-bit adders are simulated at various frequencies ranging from 1 KHz to 7 MHz at the supply voltage of 0.2V and the outputs load capacitance of 100fF. The maximum operating frequencies of the adders are shown in Fig. 7. From Fig. 7 and Table I, it can be realized that the adders with lower delays have higher maximum operating frequencies (as expected.)

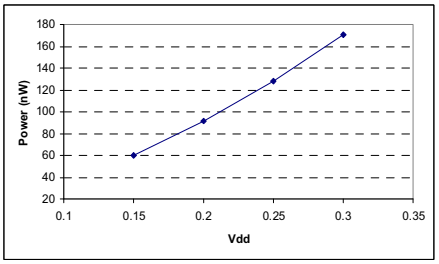
Next, the 4-bit adder which used the *bd* 1-bit full adder cell and has the least power consumption is simulated at various frequencies ranging from 1 KHz to 3MHz at the supply voltage of 0.2V and the outputs load capacitance of 100fF. The effect of changing frequency on the power consumption of this 4-bit adder is shown in Fig. 8. The major part of the power consumption in the subthreshold regime is the static power consumption which is the result of the leakage current. This current is not dependent on the frequency and, hence, the power consumption remains almost constant for a wide range of frequency. At higher frequencies, the dynamic power consumption which is proportional to the operating frequency will become dominant increasing the power consumption proportional to the frequency. For the *W/L* ratio of 6, the maximum operation frequency of 3 MHz is obtained for this adder.

The power consumption, power-delay and energy-delay product of the twelve 4-bit CRAs designed in the 65nm and 90nm technologies for the supply voltage of 0.2V and the frequency of 1MHz are compared in Fig. 9. The less the channel length is, the more the leakage current we have leading to a greater power consumption and less delay for the same supply voltage. Since the role of power consumption is more significant, the power-delay product increases too, while energy-delay product decreases due to the considerable effect of delay decrement.

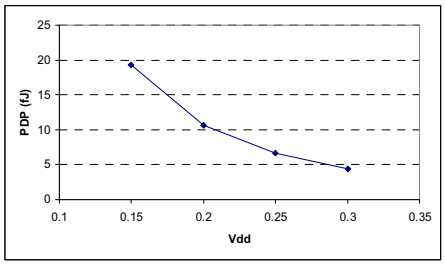
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TABLE I. SIMULATION RESULTS FOR THE TWELVE 4-BIT CARRY-RIPPLE ADDERS SORTED BY POWER

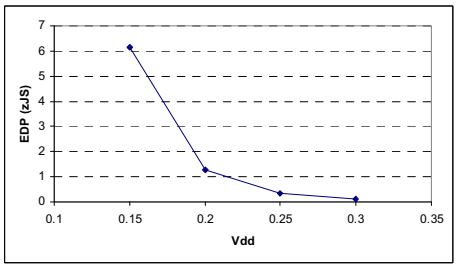
4-bit CRA	Tr. # of 1-bit full adder cell	Power (nW)	Delay (ns)	PDP (fJ)	EDP (zJs)
bd	15	91.609	117.09	10.727	1.2561
bb	14	91.632	107.04	9.8081	1.0498
cb	16	92.627	105.72	9.7927	1.0353
cd	17	92.701	106.87	9.9074	1.0588
ba	16	97.214	49.441	4.8064	0.2376
ca	18	97.638	48.580	4.7432	0.2304
ad	17	98.595	119.30	11.762	1.4032
ab	16	98.754	108.76	10.740	1.1681
bc	16	98.928	49.373	4.8844	0.2412
cc	18	99.611	51.626	5.1425	0.2655
aa	18	100.46	59.800	6.0073	0.3592
CMOS	28	101.96	54.662	5.5736	0.3047
ac	18	104.57	50.468	5.2776	0.2664
TG-CMOS	20	105.22	112.95	11.884	1.3423



(a)



(b)



(c)

Figure 6. (a) The power consumption, (b) the power-delay product and (c) the energy-delay product of the 4-bit CRA using cell *bd* versus the supply voltage at the frequency of 1MHz.

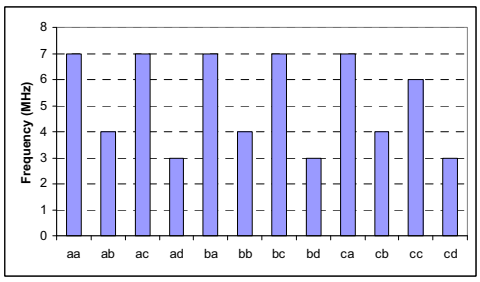


Figure 7. The maximum operating frequencies of the Carry-ripple adders

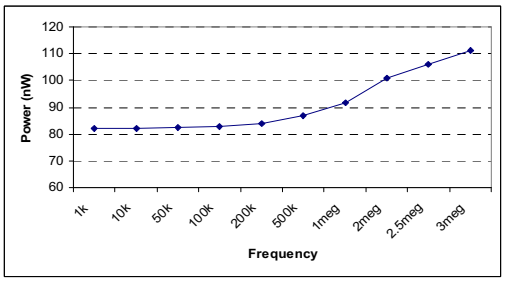
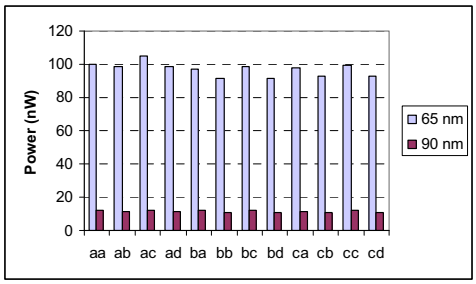
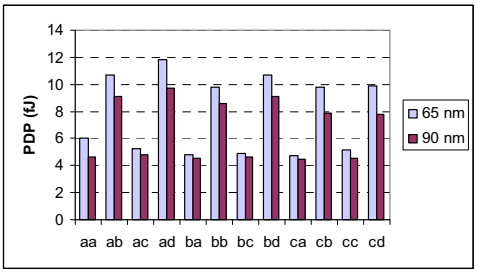


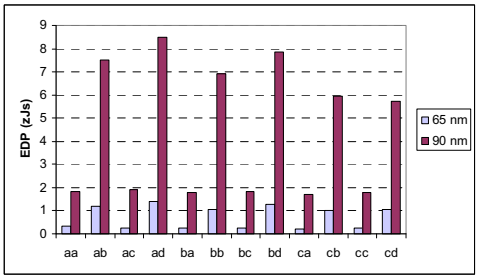
Figure 8. The power consumption versus the operating frequency for the 4-bit CRA using cell *bd*.



(a)



(b)



(c)

Figure 9. (a) The power consumption, (b) the power-delay product and (c) the energy-delay product of twelve 4-bit CRAs in 65nm and 90nm technologies.

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IV. Subthreshold 4-Bit Modified Carry Look-Ahead Adder

In this section we consider the Modified Carry Look-Ahead Adder (MCLA) [9] which has a higher speed compared to the Carry Look-Ahead adder. The structure of the adder is shown in Fig. 10. The Modified Full Adder (MFA) cell which is used to construct this adder is shown in Fig. 11.

The MCLA has remarkable increment in the area due to an excessive number of NAND gates which are used for the faster carry propagation. For solving this problem, a new Carry Look-Ahead adder is proposed [13]. In this new structure, the 4th MFA is replace by a full adder (FA) cell for reducing the area without changing speed significantly. This structure is shown in Fig. 12.

For having low power adders with acceptable speeds, one can use this MCLA in subthreshold region. For simplicity we call this new structure as MCLA in the rest of this work. This structure consist of full adder, MFA, NAND and Inverter logics. For the full adder cell, the twelve 1-bit subthreshold full adders discussed in Section II were applied. For the MFA cell, the XOR section of *Block1* of the adders is used. For the NAND and the Inverter gates, the subthreshold CMOS logics are employed.

These twelve 4-bit MCLAs have been simulated in the 65nm technology at various frequencies ranging from 1 KHz to 7 MHz with the supply voltage of 0.2V and the outputs load capacitance of 100fF. For having more realistic simulation condition, buffers are used before all the inputs. The simulation results for the twelve 4-bit adders at frequency of 1 MHz are shown in Table II. The results are sorted by the power consumption. The results show that this structure increases the power consumption as a result of more transistor count and decreases the delay of the adders because of the concurrent carry generation. This leads to the reduction in power-delay and energy-delay products. In Fig. 13, the power-delay and energy-delay products of MCLA adders are compared with those of the CRA.

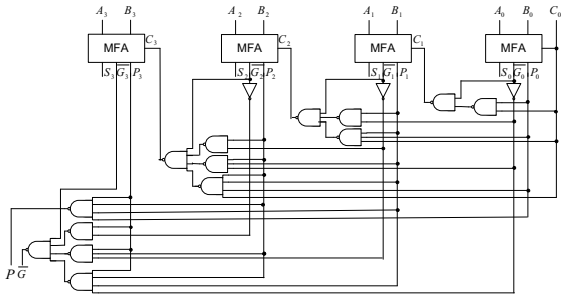


Figure 10. 4-bit Modified Carry Look-Ahead adder (MCLA) [9].

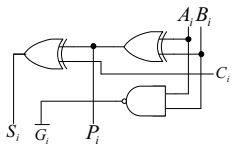


Figure 11. Modified Full Adder (MFA) cell [9].

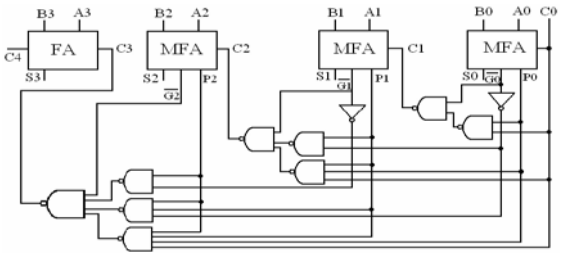
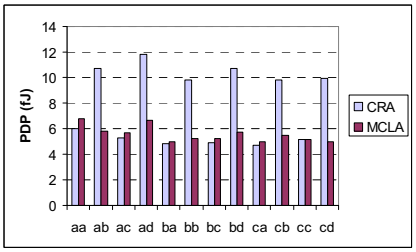


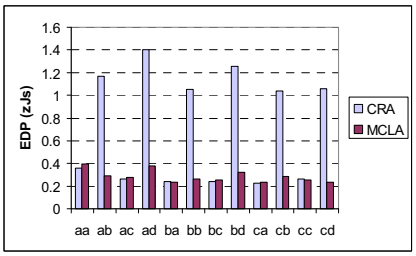
Figure 12. 4-bit Carry Look-Ahead adder proposed in [13] for reducing area.

TABLE II. Simulation Results for the 4-Bit Carry Look-Ahead Adders Sorted by Power

4-bit MCLA	Tr. # of 1-bit full adder cell	Power (nW)	Delay (ns)	PDP (fJ)	EDP (zJs)
bd	15	104.42	55.143	5.7578	0.31751
bb	14	104.44	49.907	5.2122	0.26012
cb	16	104.71	52.265	5.4727	0.28603
cd	17	104.74	47.549	4.9801	0.23680
ba	16	105.77	47.237	4.9962	0.23600
ca	18	105.98	47.128	4.9944	0.23538
cc	18	106.35	48.726	5.1821	0.25250
bc	16	106.81	49.196	5.2548	0.25851
ab	16	115.53	50.149	5.7937	0.29055
ad	17	115.53	57.305	6.6204	0.37938
aa	18	116.11	58.094	6.7451	0.39185
ac	18	117.04	48.668	5.6961	0.27721



(a)



(b)

Figure 13. (a) The power-delay product and (b) the energy-delay product of CRAs and MCLAs.

Fig. 14 shows the maximum operating frequencies of the CRA and MCLA based on the *bd* 1-bit full adder cell versus *W*/*L* ratio of the transistors. By decreasing the *W*/*L* ratio, the delay decreases and as a result the maximum operation frequency of the adders increases. The MCLA adder has a lower delay and therefore a higher maximum operation frequency. For the *W*/*L* ratio of 6 and 10, the maximum operating frequencies of MCLA with the *bd* cell are obtained as 4MHz and 6MHz, respectively. In addition to a higher

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operating frequency, this adder suffers much less from the *glitch* power due to its symmetric structure.

The power consumption, the power-delay and energy-delay products of the MCLA based on the *bd* cell versus temperature at the frequency of 1 MHz are shown in Fig. 15. It can be observed that by increasing the temperature, since the subthreshold current goes up, the power consumption increases while the delay decreases. The effect of the power increase is more significant than the delay decrease. Therefore, the power-delay and energy-delay product increase when the temperature increase.

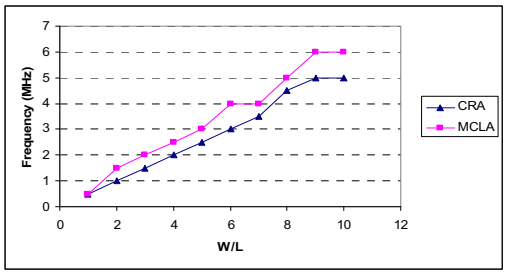
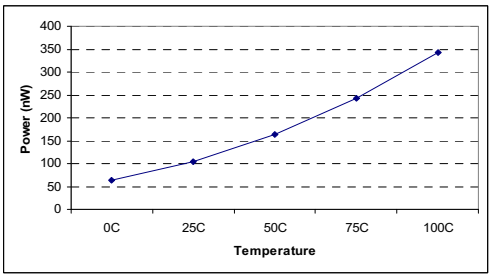
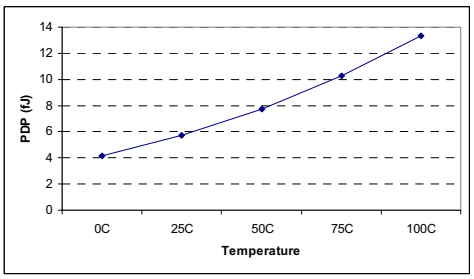


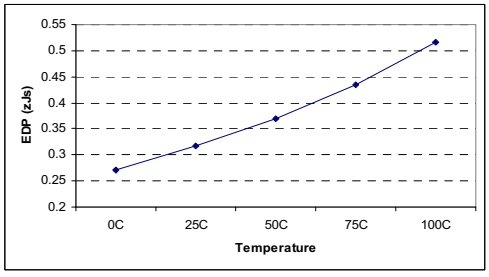
Figure 14. The maximum operation frequency of the CRA and the MCLA based on the *bd* 1-bit full adder cell versus *W*/*L* ratio of the transistors.



(a)



(b)



(c)

Figure 15. (a) The power consumption, (b) the power-delay product and (c) the energy-delay product of the MCLA using the *bd* cell versus temperature.

V. Conclusion

In this work, low power 4-bit Carry-Ripple and Modified Carry Look-Ahead adders based on subthreshold logic styles were presented. The 4-bit adders were constructed based on twelve subthreshold 1-bit full adders. To compare the performances of the adders, HSpice simulations in 65nm and 90nm standard CMOS technologies were used. The study included the effect of changing the supply voltage, sizing of the transistors, frequency, technology, and temperature. Using the simulation results the better 1-bit full adder was determined and used in the construction of the 4-bit adders. When compared to CRA, the MCLA had a lower delay, power-delay and energy-delay products.

References

[1] H. Soeleman, K. Roy, and B. C. Paul, "Robust Subthreshold Logic for Ultra-Low Power Operation," *IEEE Trans. Very Large Scale Integrat. (VLSI) Syst.*, vol. 9, no. 1, pp. 90–99, Feb. 2001.

[2] B.H.Calhoun, Al. Wang, A.Chandrakasan, "Modeling and Sizing for Minimum Energy Operation in Subthreshold Circuits," *IEEE J. Solid-State Circuits*, vol. 40, no. 9, pp. 1778–1786, Sep. 2005.

[3] Ahmed M. Shams, Tarek K. Darwish, and Magdy A. Bayoumi, "Performance analysis of low-power 1-bit CMOS full adder cells," *IEEE Trans. Very Large Scale Integrat. (VLSI) Syst.*, vol. 10, no. 1, pp. 20–29, Feb. 2002.

[4] V. Moalemi, A. Afzali-Kusha., "Subthreshold 1-Bit Full Adder Cells in sub-100 nm Technologies," IEEE Computer Society annual Symposium on VLSI, 2007, pp. 514–515.

[5] K. Granhaug and S. Aunet, "Six Subthreshold Full Adder Cells characterized in 90 nm CMOS technology," IEEE Workshop on Design and Diagnostics of Electronic Circuits and systems, 2006, pp. 25–30.

[6] H. Soeleman and K. Roy, B.Paul, "Sub-Domino Logic: Ultra-Low Power Dynamic Sub-Threshold Digital Logic," IEEE International Conference on VLSI Design, 2001, pp. 211–214.

[7] O. C. Akgun and Y. Leblebici, "Weak Inversion Performance of CMOS and DCVSPG Logic Families in Sub-300mV Range," IEEE International Symposium on Circuits and Systems (ISCAS), 2006, pp. 1251–1254.

[8] M. Lu, "Arithmetic and logic in computer systems", A John Wiley & Sons, INC.,Publication, 2004.

[9] Y. T. Pai, Y. K. Chen, "The Fastest Carry Lookahead Adder", IEEE International Workshop on Electronic Design, Test and Applications, 2004, pp. 434 – 436.

[10] B. Parhami, "Computer Arithmetic: Algorithms and Hardware Designs", Oxford University Press, 2000.

[11] Predictive Technology Model (PTM): Nanoscale Integration and Modeling (NIMO) Group at ASU [online] Available: http://www.eas.asu.edu/~ptm/.

[12] Yingtao Jiang, Abdulkarim Al-Sheraidah, Yuke Wang, Edwin Sha, and Jin-Gyun Chung "A novel multiplexer-based low-power full adder," IEEE Trans. on circuits and systems—II: Express briefs, vol. 51, no. 7, pp. 345–348, Jul. 2004.

[13] H. Thapliyal, N. Gopi, K. K. P. Kumar, "Low Power Hierarchical Multiplier and Carry Look-Ahead Architecture", IEEE International Conference on Computer Systems and Applications, Mar. 2006, pp. 88 – 92.

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